Fourier Series

G V V Sharma*

***Abstract*—This manual provides a simple introduction to Fourier Series. The final section records the measured output values from the USB power-supply experiment.**

1. Periodic Function

1.1. The unit step function is
Let

$$x(t) = A_0 \left|\sin\left(2\pi f_0 t\right)\right| \tag{1.1.1}$$

1.1 Plot $x(t)$.
1.2 Show that $x(t)$ is periodic and find its period.

2. Fourier Series

Consider $A_0 = 12$ and $f_0 = 50$ for all numerical calculations.

2.1 If

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \tag{2.1.1}$$

show that

$$c_k = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi k f_0 t}\, dt \tag{2.1.2}$$

2.2 Find c_k for (1.1.1).
2.3 Verify (1.1.1) using Python.
2.4 Show that

$$x(t) = \sum_{k=0}^{\infty} \left(a_k \cos(2\pi k f_0 t) + b_k \sin(2\pi k f_0 t)\right) \tag{2.4.1}$$

and obtain the formulae for a_k and b_k.
2.5 Find a_k and b_k for (1.1.1).
2.6 Verify (2.4.1) using Python.

*The author is with the Department of Electrical Engineering, Indian Institute of Technology Hyderabad, 502285, India. E-mail: gadepall@iith.ac.in. All content in the manuscript is released under GNU GPL. Free to use for anything.

3. Fourier Transform

3.1

$$\delta(t) = 0, \quad t \neq 0 \tag{3.1.1}$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1 \tag{3.1.2}$$

3.2 The Fourier Transform of $g(t)$ is

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t}\, dt. \tag{3.2.1}$$

3.3 Show that

$$g(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} G(f) e^{-j2\pi f t_0}. \tag{3.3.1}$$

3.4 Show that

$$G(t) \overset{\mathcal{F}}{\longleftrightarrow} g(-f). \tag{3.4.1}$$

3.5 $\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} ?$
3.6 $e^{-j2\pi f_0 t} \overset{\mathcal{F}}{\longleftrightarrow} ?$
3.7 $\cos(2\pi f_0 t) \overset{\mathcal{F}}{\longleftrightarrow} ?$
3.8 Find the Fourier Transform of $x(t)$ and plot it. Verify using Python.
3.9 Show that

$$\text{rect}\,(t) \overset{\mathcal{F}}{\longleftrightarrow} \text{sinc}\,(t)\,. \tag{3.9.1}$$

Verify using Python.
3.10 $\text{sinc}\,(t) \overset{\mathcal{F}}{\longleftrightarrow} ?$. Verify using Python.

4. Filter

4.1 Find $H(f)$ which transforms $x(t)$ to DC 5 V.
4.2 Find $h(t)$.
4.3 Verify your result using convolution.

5. Filter Design

5.1 Design a Butterworth filter for $H(f)$.
5.2 Design a Chebyshev filter for $H(f)$.
5.3 Design a circuit for your Butterworth filter.
5.4 Design a circuit for your Chebyshev filter.

TABLE I: Input and transformer values

Quantity	Value
Primary RMS voltage	220 V
Mains frequency	50 Hz
Primary peak voltage (calculated)	311.1 V
Measured secondary RMS voltage	10.53 V
Secondary peak voltage (calculated)	14.89 V
Approximate turns ratio	20.9 : 1

TABLE II: Bridge rectifier values from the previous draft

Quantity	Value
Measured secondary RMS voltage	10.53 V
Secondary peak voltage	14.89 V
Assumed forward drop per diode	0.7 V
Bridge peak voltage	13.49 V
Rectified average voltage	8.59 V
Rectified RMS voltage	9.54 V
Input frequency	50 Hz
Full-wave ripple frequency	100 Hz
Ripple period	10 ms

6. Circuit Input and Transformer

The circuit calculations in the earlier draft use a nominal primary supply of 220 V RMS at 50 Hz and a measured secondary voltage of 10.53 V RMS. The transformer reduces the AC voltage and provides galvanic isolation.

For a sinusoidal waveform, the peak voltage is $\sqrt{2}$ times the RMS voltage. The ideal transformer voltage ratio is equal to its turns ratio. The values above are the values used in the previous draft.

The reference circuit document separately describes a 12–0–12 V centre-tapped transformer, a 12 V RMS winding output, and an approximately 18 V peak level. These reference values are not the same measurement as the 10.53 V RMS secondary used for the calculations above.

7. Full-Wave Bridge Rectifier

The bridge rectifier uses four diodes. Two diodes conduct during each half-cycle, so the current through the load remains in the same direction. The rectified waveform has twice the input frequency.

For the previous draft's first-order calculation, the forward drop of each conducting silicon diode is taken as 0.7 V. The calculated bridge peak is 13.49 V; the average and RMS values of the idealised rectified waveform are 8.59 V and 9.54 V.

The earlier component comparison listed the 1N4007 diode (1 A average forward-current class and 1000 V repetitive peak reverse-voltage rating) and the 1N5408 diode (3 A average forward-current class).

8. Smoothing Capacitor

The smoothing capacitor charges near the peaks of the rectified waveform and supplies the load between peaks. This reduces the voltage variation at the rectifier output.

The previous draft uses a 2200 μF capacitor for its main ripple calculation. For a full-wave rectifier, the approximate peak-to-peak ripple relation used was

$$V_{r(pp)} \approx \frac{I_L}{f_r C}.$$

At $f_r = 100$ Hz and $C = 2200\,\mu$F, this gives

$$V_{r(pp)} \approx 4.545 I_L \text{ V},$$

where I_L is in amperes. Using the calculated peak of 13.49 V, the corresponding approximate average is

$$V_{\mathrm{DC}} \approx 13.49 - 2.273 I_L \text{ V}.$$

These are the previous draft's calculated relations, not additional measurements.

The separate reference circuit document lists a 100 μF filter capacitor and reports an approximately 18 V DC level for its own 12 V RMS / approximately 18 V peak example. The 100 μF and 2200 μF values belong to different records and are kept separate here.

9. Regulator and Buck Converter

The earlier draft compared a 7805 linear regulator with an LM2596 step-down switching regulator. These are alternative regulation approaches.

A. 7805 Linear Regulator

The 7805 is a fixed-output linear regulator with a nominal 5 V output. The previous draft records approximately 2 V dropout at a stated 1 A test condition for the TI LM340/LM7805 family. The actual dropout depends on the specific device, load, and temperature.

The regulator's approximate power dissipation is

$$P_D \approx (V_{\mathrm{IN}} - V_{\mathrm{OUT}}) I_{\mathrm{OUT}}.$$

TABLE III: Regulator comparison

Feature	7805	LM2596
Method	Linear	Switching buck
Output	Fixed 5 V	Fixed or adjustable
Main consideration	Headroom and heat	Ripple, layout, and heat
Key previous-draft values	About 2 V dropout at stated 1 A test	4.5–40 V input; up to 3 A; 150 kHz

The 7805 family documentation also describes current limiting, thermal shutdown, and safe-operating-area protection. A current limit of approximately 2.4 A was noted in the earlier reference material as a protection figure, not as a continuous output-current rating. The previous draft also noted that capabilities above 1.5 A depend on the device variant and thermal conditions.

B. LM2596 Buck Converter

The LM2596 is a switching step-down regulator. The previous draft records the following family specifications: input range 4.5–40 V, output-current capability up to 3 A under specified conditions, and a 150 kHz oscillator frequency. The current capability of a complete module also depends on its inductor, diode, capacitors, circuit board, and cooling.

For an ideal buck converter, the duty ratio is

$$D = \frac{V_{\mathrm{OUT}}}{V_{\mathrm{IN}}}.$$

The earlier draft's 12 V to 5 V example corresponds to an ideal duty ratio of approximately 41.7%. This is a reference calculation, not a measured operating point of the reported USB output.

10. USB Output Port

The USB output includes a positive supply contact (VBUS), ground (GND), and interface contacts. In USB 2.0, D+ and D- carry data and may also participate in legacy charging-source detection. USB Type-C uses configuration-channel (CC) pins to advertise available current. USB Power Delivery (USB-PD) requires a compatible controller and negotiation; a fixed 5 V regulator alone does not provide USB-PD.

**Applicable USB enumeration and configuration rules apply.*

TABLE IV: USB charging current values listed in the previous draft

Port or mode	Current value
USB 2.0 SDP	Up to 500 mA*
USB 3.x SDP	Up to 900 mA*
BC1.2 CDP	Up to 1.5 A
BC1.2 DCP	Up to 1.5 A
USB Type-C current advertisement	1.5 A or 3 A at 5 V

TABLE V: Legacy D+/D- signatures recorded in the reference

D+	D-	Reported current
2.0 V	2.0 V	Approx. 0.5 A
2.0 V	2.7 V	Approx. 1.0 A
2.7 V	2.0 V	Approx. 2.1 A
2.7 V	2.7 V	Approx. 2.4 A

TABLE VI: Measured USB output values

Parameter	Measured value
Average output voltage	4.94 V
Average output current	0.60 A
Additional instrument display	16 mAh

The previous draft also included these legacy proprietary D+/D- voltage signatures. They are retained as reference values and are not universal USB charging rules.

11. Measured USB Output

The USB output was measured using a voltmeter and an ammeter. The recorded average voltage was approximately 4.94 V, and the recorded average current was approximately 0.60 A. The instrument also displayed 16 mAh; this is recorded separately because mAh is a charge-capacity unit, not a current unit.